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Effects of Short-Term Calcifediol Supplementation on Leukocyte Cytokine Production in Healthy Dogs: A Randomized, Double-Blinded, Placebo-Controlled Crossover Trial

Jared A. Jaffey[1] | Rachael Kreisler[2] | Robert C. Backus[3] | Danielle Gordon[2] | Lauren Chittick[1]

[1]Department of Specialty Medicine, Midwestern University College of Veterinary Medicine, Glendale, Arizona, USA | [2]Department of Pathology and Population Medicine, Midwestern University College of Veterinary Medicine, Glendale, Arizona, USA | [3]Department of Veterinary Medicine and Surgery, Veterinary Health Center, University of Missouri, Columbia, Missouri, USA

Correspondence: Jared A. Jaffey (jjaffe@midwestern.edu)

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ABSTRACT

Background: Vitamin D modulates the immune response in many species, including dogs. To date, research investigating the immunological effects of vitamin D in dogs is limited to in vitro studies.

Objectives: Provide PO calcifediol supplementation to healthy dogs to evaluate its tolerability and assess its effect on leukocyte production of tumor necrosis factor (TNF)-α, interleukin (IL)-6, and IL-10.

Animals: Eleven healthy client-owned dogs with serum 25-hydroxyvitamin-D_3 (25(OH)D_3) concentrations ≤ 30 ng/mL.

Methods: Prospective, randomized, double-blinded, placebo-controlled crossover study. Dogs were randomized to receive calcifediol at 2.3 μg/kg$^{0.75}$ (low-dose), 4.6 μg/kg$^{0.75}$ (high-dose), or placebo for 7 days and crossed over to a different treatment arm after 28-day washout periods. Serum 25(OH)D_3 was measured using a modified high-performance liquid chromatography method. Whole blood cultures were performed by incubating blood with lipopolysaccharide (LPS), lipoteichoic acid (LTA), or phosphate-buffered saline (PBS) for 24 h, and TNF-α, IL-6, and IL-10 were measured in supernatant using a canine-specific multiplex assay.

Results: Both low-dose (median, interquartile range [IQR]; 62 ng/mL, 59–82) and high-dose (93 ng/mL, 80–113) PO calcifediol supplementation increased serum 25(OH)D_3 concentrations from baseline (22.2 ng/mL, 20.3–29.3; both $p < 0.0001$). Low-dose and high-dose calcifediol supplementation decreased LPS-stimulated IL-6 by 197 pg/mL (95% confidence interval [CI]: −369.0 to −24.0; $p = 0.03$) and 182 pg/mL (95% CI: −355.0 to −8.0; $p = 0.04$), respectively. Then LPS-stimulated IL-6 concentrations decreased by 2 pg/mL for every 1 ng/mL increase in serum 25(OH)D_3 concentration (95% CI: −3.0 to −0.1; $p = 0.03$).

Conclusion and Clinical Importance: Calcifediol supplementation for 7 days was well-tolerated and decreased LPS-stimulated IL-6 concentrations. Oral calcifediol supplementation may have anti-inflammatory effects in dogs.

Abbreviations: 25(OH)D_3, 25-hydroxyvitamin D; BUN, blood urea nitrogen; CI, confidence interval; CKD, chronic kidney disease; FGF-23, fibroblast growth factor-23; HPLC, high-performance liquid chromatography; iCa, ionized calcium; IL, interleukin; IQR, interquartile range; LC–MS/MS, liquid chromatography–tandem mass spectrometry; LOWESS, locally-weighted scatterplot smoothing; LPS, lipopolysaccharide; LTA, lipoteichoic acid; MELR, mixed effects linear regression; PBS, phosphate-buffered saline; PTH, parathyroid hormone; RI, reference interval; RPMI, Roswell Park Memorial Institute; TNF-α, tumor necrosis factor-α; VDR, vitamin D receptor.

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1 | Introduction

Vitamin D is best known for its role in calcium and phosphorus homeostasis and musculoskeletal health. However, research over the last several decades in people and more recently in domestic animals has shown that the biological importance of vitamin D extends beyond these basic physiologic processes. Genomic studies in human medicine have shown that the vitamin D receptor (VDR) has nearly ubiquitous expression in tissues and cells (www.proteinatlas.org/ENSG00000111424-VDR/tissue, accessed on 1 December 2024) and binding with 1,25-dihydroxyvitamin D (calcitriol), the biologically active form of vitamin D, regulates immunity, cell growth, and energy metabolism [1–3]. A small study that included 6 dogs and 13 tissue types identified VDR expression of variable magnitude in kidneys, skin, gastrointestinal tract, lymph node, heart, liver, ovary, and lung [4]. Further emphasizing the integral role of vitamin D is the fact that many types of cells beyond renal epithelium are capable of converting 25-hydroxyvitamin-D_3 (25(OH)D_3; calcifediol), the primary circulating form of vitamin D, to calcitriol [5].

Vitamin D plays an integral role in various aspects of immunomodulation across many species, including dogs, chickens, cattle, and people [6–15]. The attenuation of pro-inflammatory cytokine production is one of the most studied modulatory effects of vitamin D that is conserved in both dogs and human patients [6–10, 16, 17]. One of several in vitro studies in dogs highlighted this effect because antigen-stimulated production of tumor necrosis factor (TNF)-α was blunted by calcitriol in a concentration-dependent manner [7]. Another observational study in dogs found an inverse association between serum 25(OH)D_3 concentrations and antigen-stimulated TNF-α concentrations. Specifically, as serum 25(OH)D_3 concentrations increased, the capacity for TNF-α production decreased [18]. Chronic unabated or exaggerated systemic inflammation can lead to a variety of potential health consequences in dogs and people [19, 20]. Therefore, utilization of supplemental vitamin D to mitigate inflammation has intriguing potential as an adjunctive treatment.

Oral supplementation with various analogues of vitamin D in people decreases markers of systemic inflammation, and several studies have shown its clinical utility as an adjunctive treatment in several disorders [21–29]. Clinical trials investigating outcome assessments with PO vitamin D supplementation in dogs have been limited to chronic kidney disease (CKD) and atopic dermatitis [30, 31]. No studies have evaluated the immunologic effects of PO vitamin D supplementation in dogs.

Given the documented associations between vitamin D and inflammatory cytokine production in dogs, we sought to investigate whether short-term PO calcifediol administration affected leukocyte cytokine production ex vivo and if immune responses were associated with serum 25(OH)D_3 concentrations in healthy dogs with vitamin D insufficiency. A secondary aim was to determine whether short-term PO calcifediol administration resulted in adverse events or disturbances in blood urea nitrogen (BUN), creatinine, or ionized calcium (iCa) concentrations. We hypothesized that calcifediol administration would have anti-inflammatory effects on the cytokine milieu, with changes being dose dependent and associations between serum 25(OH)D_3 concentrations and cytokine production would be identified.

2 | Materials and Methods

2.1 | Criteria for Selection of Cases

Healthy dogs owned by students, staff, and faculty at Midwestern University College of Veterinary Medicine with serum 25(OH)D_3 concentrations ≤ 30 ng/mL (vitamin D insufficient) that were > 1 year of age were eligible for inclusion. A consensus on optimal vitamin D status in a population of healthy dogs has not been reached. Therefore, vitamin D insufficiency was defined as serum 25(OH)D_3 concentrations ≤ 30 ng/mL, which is a commonly used threshold in human patients [32, 33]. Healthy status was based on review of clinical history, physical examination findings, CBC, serum biochemical panel, and urinalysis (Antech Diagnostics, Irvine, CA) by a single board-certified internist (JAJ). Dogs could not have a body condition score > 6/9, have had an illness within 3 months, or received any vaccinations within 1 month of enrollment. Monthly parasiticides were the only permissible medication. Diet information was recorded for all dogs, and each continued to eat its standard type and amount of diet throughout the study period. Diets were not evaluated for vitamin D supplementation. Dogs were excluded if they did not adhere to the pre-defined assessment schedule, developed a clinically relevant illness, or a deviation from the standard feeding regimen occurred. The study protocol was approved by the Midwestern University Animal Care and Use Committee (protocol: 3017) with written owner consent. Dogs were enrolled from December 2020 to January 2021.

2.2 | Study Design

A randomized, double-blinded, placebo-controlled crossover study was performed. Dogs were screened initially with a review of clinical history, physical examination findings, and serum 25(OH)D_3 concentrations. A delay in turnaround time occurred related to the COVID-19 pandemic for the initial screening of serum 25(OH)D_3 concentrations. Dogs returned within a 6-week time frame to complete enrollment, which consisted of repeating a review of clinical history, physical examination, and evaluation of CBC, serum biochemical panel, and urinalysis. After inclusion in the study, dogs were randomized by block randomization to receive calcifediol at 2.3 μg/kg$^{0.75}$ (low-dose), calcifediol at 4.6 μg/kg$^{0.75}$ (high-dose), or placebo in one of 6 possible sequences, with each sequence assigned 2 dogs (Figure 1). Dogs were in each treatment arm for 7 days and crossed over to the next treatment in their sequence after 28-day washout periods (Figure 1). Dogs were fasted for 12 h before each blood collection. The lower calcifediol dose was extrapolated from two studies that each utilized healthy dogs and dogs with CKD, respectively [31, 34]. The rationale for the high dose was to double the low dose to evaluate cytokine production across a broader range of serum 25(OH)D_3 concentrations. A 28-day washout period was

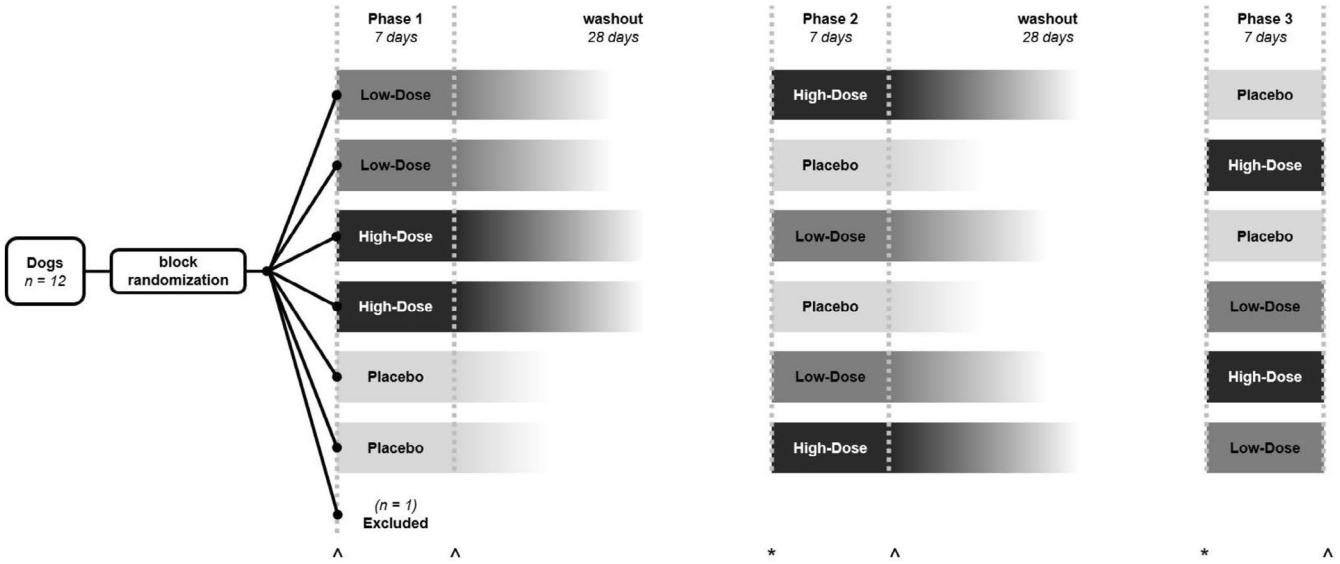

FIGURE 1 | Illustrates study design and annotations for blood sampling. Twelve dogs were included in the study and were randomized via block randomization to receive calcifediol at 2.3 μg/kg$^{0.75}$ (low-dose), at 4.6 μg/kg$^{0.75}$ (high-dose), or placebo in one of 6 possible sequences, with each sequence assigned 2 dogs. Dogs were in each treatment arm for 7 days and crossed over to the next treatment in their sequence following 28-day washout periods. One dog was excluded after randomization but before initiating the first treatment arm. Circumflex symbol (^) represents blood sampling for blood gas analysis, immunological experiments, and serum 25-hydroxyvitamin-D_3 [25(OH)D_3] measurement. Asterisk symbol (*) represents blood sampling at the end of each washout period for serum 25(OH)D_3 quantification.

chosen based on a previously calculated plasma half-life of 1.8 weeks for 25(OH)D_3 in dogs [34]. Short-term adverse events that occurred throughout the study period were recorded. An adverse event was defined as any observation, unwanted experience, or reaction that was unfavorable, unintended, and occurred after the initiation of the study. A power analysis was not conducted because of a lack of published data in this area. Instead, a sample size of 12 dogs in a crossover design was chosen based on visual assessment of in vitro data on calcitriol's effects on leukocyte cytokine production in dogs, as well as practical considerations related to available resources.

At baseline (i.e., 1 day before the start of the first treatment arm), and at the end of each treatment arm, blood samples were collected for blood gas analysis, 25(OH)D_3 measurement, and immunologic experiments (Figure 1). Blood also was collected at the end of each of the 2 washout periods for serum 25(OH)D_3 quantification. Blood was collected into lithium heparin anticoagulated blood tubes for blood gas analysis and immunologic experiments. Whole blood was analyzed using a veterinary benchtop blood gas analyzer (Stat Profile Prime Plus Vet, Nova Biomedical, MA) within 5 min of collection. Throughout the clinical trial, the blood gas analyzer underwent quality control daily and routine maintenance as recommended by the manufacturer. Variables recorded from each blood gas analysis included iCa (reference interval [RI]: 1.16–1.40 mmol/L), creatinine (RI: 0.6–1.3 mg/dL), and BUN (RI: 9.0–26.0 mg/dL) concentrations. Immunologic experiments, as outlined below, were initiated within 1 h of blood sample collection. Blood for 25(OH)D_3 quantification was transferred into serum separator blood tubes, allowed to clot, and then centrifuged with serum transferred to freezer-resistant conical microcentrifuge tubes and stored at −80°C for batch analysis.

2.3 | Calcifediol and Placebo Capsule Preparation

Capsules were compounded by a commercial veterinary pharmacy (Pet Health Pharmacy, AZ) following compounding standards set by United States Pharmacopeia chapters 795 and 1163. Calcifediol capsules were filled with the same active pharmaceutical ingredient, bulk powder (HyD, DSM-Firmenich AG LLC, NK). The use of HyD for calcifediol supplementation was chosen for its demonstrated safety and effectiveness in dogs [35]. Proprietary calcifediol (HyD) is a supplement that has been used in poultry feeds for more than 20 years and more recently in swine. Placebo capsules were compounded using anhydrous lactose powder. Capsules for each treatment appeared visually the same with the exception that they were color coded for the purpose of blinding, and this designation remained confidential from investigators and dog owners until completion of the study. Compliance was assessed at the end of each treatment arm with the return of the drug vials to study personnel.

2.4 | Calcifediol Capsule Analysis

Twenty-two capsules were selected for analysis (low-dose, $n = 11$; high-dose, $n = 11$). Each of the 11 dogs had 1 capsule each of both calcifediol doses analyzed. The capsule selected for analysis was chosen at the end of the study based on the pre-planned surplus capsule for each treatment from each dog. Specifically, it was decided a priori that an additional capsule for all three treatments would be produced for the purpose of analyzing calcifediol concentrations in the compounded capsules. Calcifediol capsules were analyzed by DSM-Firmenich using previously reported methods for quantification of calcifediol in feeds and premixes containing HyD [36]. Briefly, capsule contents and deuterated internal standard (d6-25-hydroxyvitamin D_3) were combined

in an alcoholic aqueous solution, saponified, and extracted into tert-butyl methyl ether. The organic phase was separated and dried with resulting residue reconstituted and fractionated by reverse-phase high-performance liquid chromatography (HPLC). Eluted 25(OH)D_3 and internal standard were quantified using liquid chromatography–tandem mass spectrometry (LC–MS/MS), thereby allowing calculation of the initial calcifediol content of capsules. The analytical method was reported by DSM to be valid for feeds and premixes containing calcifediol concentrations > 10 μg/kg and < 3000 μg/kg for which it was deemed sufficient in selectivity, linearity, extraction efficiency, accuracy, and precision. Laboratory personnel were blinded to the calcifediol target dose in each capsule.

2.5 | Serum 25(OH)D_3 Concentrations

Serum concentrations of 25(OH)D_3 were determined in thawed samples using previously reported HPLC methods [37]. The lower limit of quantification was 3.0 ng/mL.

2.6 | Whole Blood Cultures and Cytokine Measurements

Lithium-heparin anticoagulated blood was diluted 1:2 with Roswell Park Memorial Institute (RPMI) 1640 culture medium (Thermo Fisher Scientific, CA) containing 200 U of penicillin/mL and 200 mg of streptomycin/mL. The blood-RPMI mixture then was transferred to 96-well plates and stimulated with lipopolysaccharide (LPS) from *Escherichia coli* O127:B8 (final concentration, 100 ng/mL, Sigma Aldrich, MO), lipoteichoic acid (LTA) from *Streptococcus faecalis* (final concentration, 1 μg/mL, Sigma Aldrich, St. Louis, MO), or phosphate-buffered saline (PBS) control. Plates were incubated for 24 h at 37°C in 5% CO_2 in the dark. After incubation, plates were centrifuged (400×*g* for 7 min) at 21°C as previously described [6]. The supernatant was collected and stored at −80°C for batch analysis. For analysis, samples were thawed, and then TNF-α, IL-6, and IL-10 were measured in supernatant using a cytokine-specific multiplex bead-based assay designed for dogs (Milliplex MAP canine cytokine-chemokine panel, EMD Millipore Corp, MA). The median fluorescence intensity and cytokine concentration in each sample were measured in duplicate using appropriate controls and associated data analysis software (Milliplex Analyst version 5.1, EMD Millipore Corp, MA). The lower limit of detection for TNF-α and IL-6 was 48.8 pg/mL, and for IL-10 was 195 pg/mL.

3 | Statistical Analysis

Statistical analysis was performed using commercial software (Stata Statistical Software version 18, StataCorp LLC, TX). Data were assessed for normality using tests of skewness and kurtosis, and reported as mean and SD for normally distributed data and median and interquartile range (IQR; reported as quartiles 1 and 3) for non-normally distributed data. When the measured cytokine concentrations fell below the lower limit of detection, data were recorded at the lower limit of detection for statistical purposes. Comparison of serum 25(OH)D_3 concentrations at the time of initial screening to baseline measurements was made using a Wilcoxon signed-rank test. Correlation between target and actual calcifediol dose was assessed using Spearman's rank correlation coefficient (r_s). Scatterplots were used to visualize the relationship between 25(OH)D_3 and multiple physiologic variables, including iCa, BUN, and creatinine. Scatterplots also were used to assess the relationship between 25(OH)D_3 and cytokines (TNF-α, IL-6, and IL-10), stimulated by PBS, LPS, and LTA. Multivariable mixed effects linear regression (MELR) with dog as random effect was used to model associations between serum 25(OH)D_3 and treatment group while controlling for potential carryover effects from the preceding treatment. Similarly, MELR clustered on dog was used to model serum 25(OH)D_3 and cytokine concentrations. Competing models, including those with interaction terms, were compared using Akaike information criterion and Bayesian information criterion and validated from inspection of the residuals. Adequacy of the washout period was assessed using a Kruskal–Wallis test comparing baseline to washout periods 1 and 2, followed by a Dunn's test for pairwise comparison. $p < 0.05$ was considered significant.

4 | Results

4.1 | Dogs

Twenty-three dogs initially were screened for enrollment. The median serum 25(OH)D_3 concentration was 20.5 ng/mL (IQR, range; 17.4–27.2, 14.7–42.7 ng/mL). As explained earlier, based on the threshold for vitamin D insufficiency used in humans (≤ 30 ng/mL), 83% (19/23) of the dogs were vitamin D insufficient, and the median serum 25(OH)D_3 concentration in this subgroup was 19.5 ng/mL (IQR, range; 16.9–23.3, 14.7–27.3 ng/mL). Twelve dogs were randomly selected from the pool of eligible dogs and subsequently were enrolled, but 1 dog was excluded after randomization and shortly before the first treatment arm, leaving 11 dogs in the study. The dog was excluded because the owner relocated. The mean age and weight were 5.0 years (SD, 2.1) and 15.5 kgs (SD, 10.8), respectively. Most dogs were mixed breed dogs (91%, 10/11) with 1 Chihuahua. There were seven spayed female and four neutered male dogs. Dogs were fed a commercially available pet food. All dogs completed the three treatment arms of the study without any reported adverse events. Complete compliance was verified in all dogs.

4.2 | Serum 25(OH)D_3 Concentrations

The median serum 25(OH)D_3 concentrations at the time of initial screening was 18.3 ng/mL (IQR, range; 16.7–20.5, 14.7–27.3 ng/mL). A significant increase in serum 25(OH)D_3 concentrations was observed from the time dogs were screened to baseline (i.e., 1 day before start of treatment arm 1; median, IQR; 22.2 ng/mL, 20.3–29.3; $p = 0.01$). Two dogs had serum 25(OH)D_3 concentrations > 30 ng/mL at baseline (32.2 ng/mL and 37.9 ng/mL) but had concentrations below the predefined threshold for vitamin D insufficiency when initially screened.

The median serum 25(OH)D_3 concentrations after low-dose and high-dose calcifediol were 62.0 ng/mL (IQR, 59.0–82.0) and 93 ng/mL (IQR, 80–113), respectively (Figure 2). The median percentage increase in serum 25(OH)D_3 after calcifediol

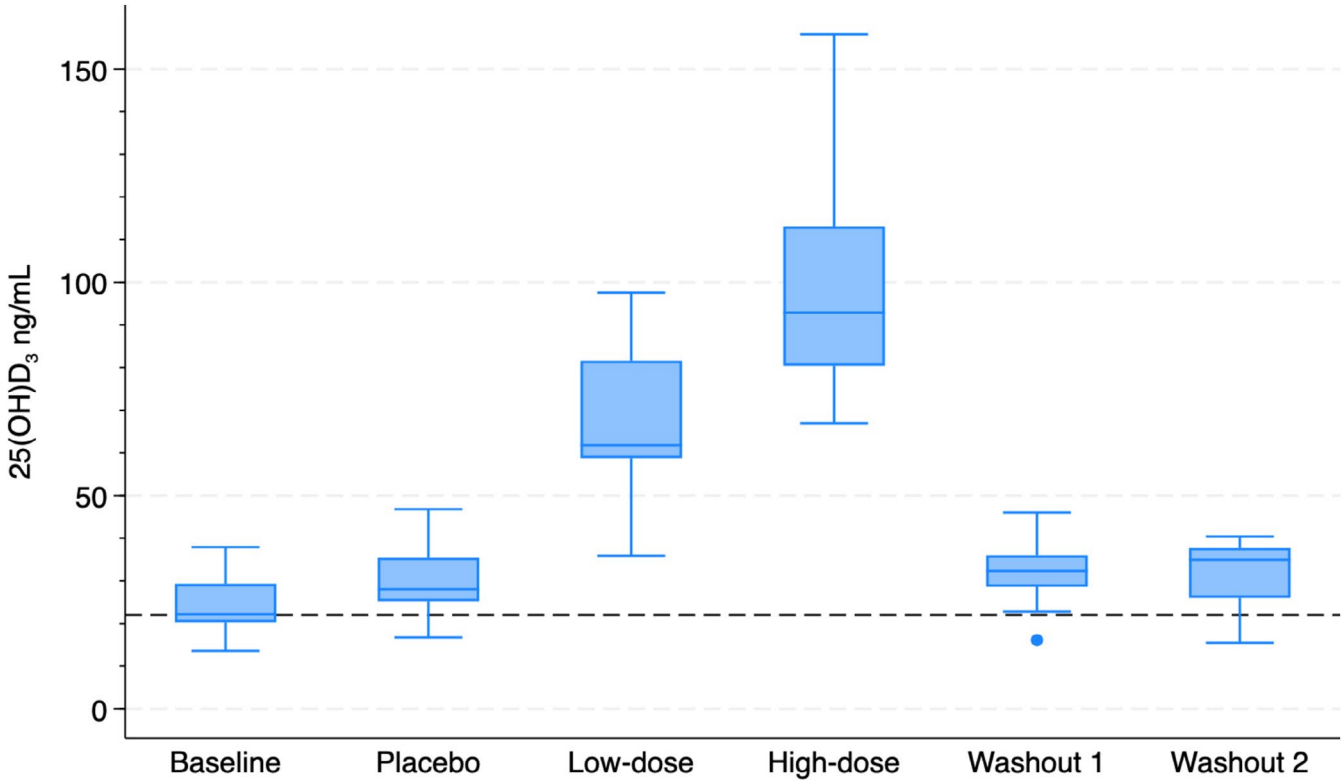

FIGURE 2 | Box-and-whisker plots comparing serum 25-hydroxyvitamin-D_3 [25(OH)D_3] concentrations at baseline (i.e., 1 day before treatment arm 1), after each of the 3 treatment arms, and after each washout period for 11 dogs. The central box represents the interquartile range, with the lower and upper edges corresponding to quartile 1 and quartile 3, respectively. The horizonal line inside the box indicates the median. The whiskers extend to the smallest and largest values within 1.5× the IQR. Outliers are represented as individual points beyond the whiskers. The horizontal dashed line represents the median at baseline (22 ng/mL) for direct comparison.

was administered as compared to baseline at the low dose was 162.4% (IQR, 107.5–302.0) and at the high dose was 313.9% (IQR, 265.6–457.1). In a MELR model controlling for the preceding treatment (carryover effect), both calcifediol doses increased serum 25(OH)D_3 concentrations as compared to baseline (both $p < 0.0001$), whereas placebo was not different ($p = 0.55$; Figure 2). Low-dose calcifediol increased 25(OH)D_3 concentrations by 36 ng/mL, whereas the high dose increased concentrations by 71 ng/mL. Additional increases of 13 ng/mL and 16 ng/mL were observed for all treatments if the preceding treatment was low dose ($p = 0.05$) or high dose ($p = 0.02$), respectively.

For placebo, washout period 1 and washout period 2, the median 25(OH)D_3 concentrations were 28 ng/mL (IQR, 25–35), 32 ng/mL (IQR, 29–26) and 35 ng/mL (IQR, 26–38), respectively. A Kruskal–Wallis test comparing 25(OH)D_3 concentrations at baseline, washout 1 and washout 2 identified a significant difference between baseline and both washout 1 ($p = 0.02$) and washout 2 ($p = 0.008$; Figure 2). No significant difference was found between washout periods 1 and 2 ($p = 0.35$).

4.3 | Effect of Serum 25(OH)D_3 Concentration and Calcifediol Dose on Cytokine Concentration

The LPS-stimulated IL-6 concentrations decreased by 2 pg/mL for every 1 ng/mL increase in serum 25(OH)D_3 concentration (95% CI: −3.0 to −0.1; $p = 0.03$; Figure 3). When considered by calcifediol dose, LPS-stimulated IL-6 concentrations decreased by 197 pg/mL for low dose (95% CI: −369.0 to −24.0; $p = 0.03$) and 182 pg/mL for high dose (95% CI: −355.0 to −8.0; $p = 0.04$) as compared with baseline (Figure 4). Although a decrease of 131 pg/mL occurred for placebo as compared to baseline, this difference was not significant (95% CI: −273.2 to 11.8; $p = 0.07$). Figure S2 illustrates longitudinal changes in individual dog data for LPS-stimulated IL-6 concentrations. Visual inspection of the relationship between 25(OH)D_3 and LPS-stimulated TNF-α concentrations (Figure 5) showed a linear decrease that was not significant based on MELR ($p = 0.31$). No visual relationship was found between serum 25(OH)D_3 and the remaining cytokine concentrations under any condition (Figure S1) nor were there any between-treatment differences (Table 1).

4.4 | Blood Chemistry Variables (iCa, BUN, Creatinine) and Serum 25(OH)D_3

The mean iCa, BUN, and creatinine concentrations at baseline were 1.32 mmol/L (SD, 0.05), 17.9 mg/dL (SD, 2.6), and 1.2 mg/dL (SD, 0.19), respectively. Three dogs had creatinine concentrations at baseline mildly above the upper limit of the RI for the blood gas analyzer. Two dogs each had a whole blood concentration of 1.4 mg/dL, and the other dog had a creatinine concentration of 1.5 mg/dL. The serum biochemical panel from the commercial reference laboratory for these 3 dogs indicated creatinine concentrations within the RI, with results of 0.9, 1.1, and 1.5 mg/dL.

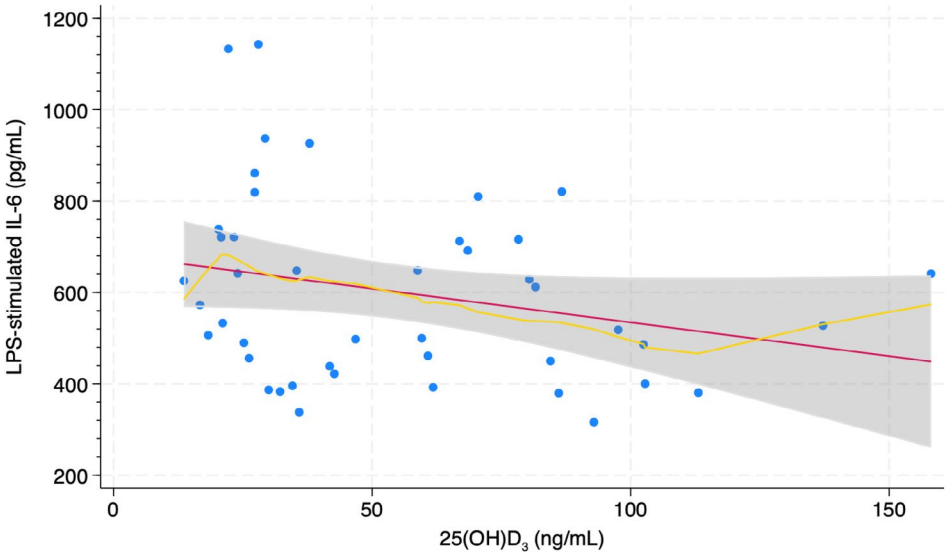

FIGURE 3 | Scatterplot of lipopolysaccharide (LPS)-stimulated interleukin (IL)-6 by 25-hydroxyvitamin-D_3 [25(OH)D_3] overlaid with a linear best fit line (magenta line) with 95% confidence intervals. Linear fit confirmed with LOWESS (yellow line). Four data points from 11 dogs, for a total of 44 data points.

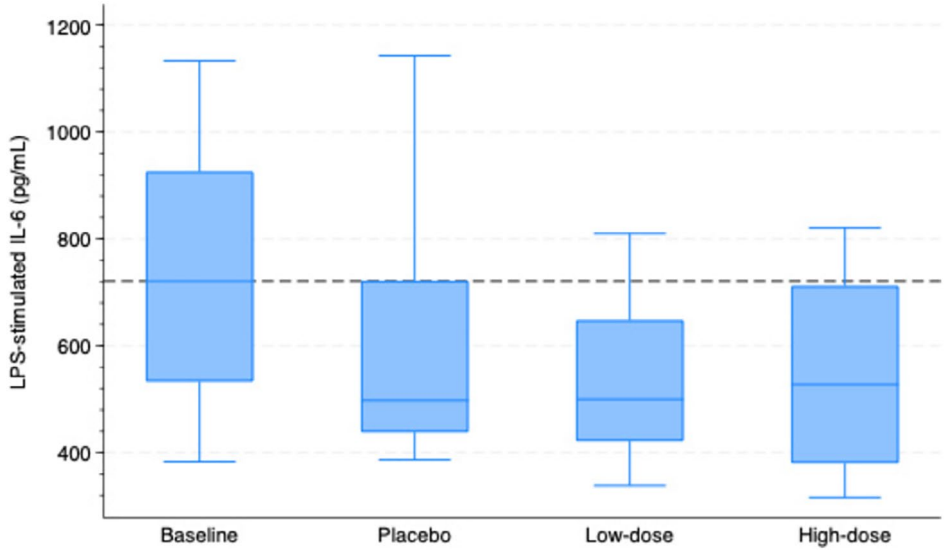

FIGURE 4 | Box-and-Whisker plots comparing the effect of treatment (i.e., placebo, low-dose calcifediol, and high-dose calcifediol) on lipopolysaccharide (LPS)-stimulated interleukin (IL)-6 production. The central box represents the interquartile range, with the lower and upper edges corresponding to quartile 1 and quartile 3, respectively. The horizonal line inside the box indicates the median. The whiskers extend to the smallest and largest values within 1.5× the IQR. Outliers are represented as individual points beyond the whiskers. The horizontal dashed line represents the median at baseline (721 pg/mL) for direct comparison.

Seven creatinine concentration results (21%, 7/33) from 4 dogs throughout the study period were mildly above the upper limit of the RI. The median whole blood creatinine concentration from these seven results was 1.5 mg/dL (IQR, 1.4–1.5). One dog developed a mild increase in iCa concentration after treatment with placebo (1.43 mmol/L) but not after calcifediol administration. No other dogs developed ionized hypercalcemia at any point. Blood urea nitrogen concentrations remained within the RI throughout the study period in all dogs.

Visual inspection of the scatterplots of the relationship between serum 25(OH)D_3 and the physiologic variables of iCa, BUN, and creatinine overlaid with a locally weighted scatterplot smoothing (LOWESS) curve did not identify a relationship (Figure S3), which was confirmed using MELR (all $p > 0.5$; Table S1).

4.5 | Calcifediol Capsule Analysis

A significant correlation was found between the target calcifediol dose in the compounded capsules and actual measured calcifediol concentration ($r_s = 0.99$; $p < 0.001$; $n = 22$). The median delivered calcifediol dose deviated by −12% (IQR, range; −18.1 to −2.6; −30% to 15%) from the intended target dose.

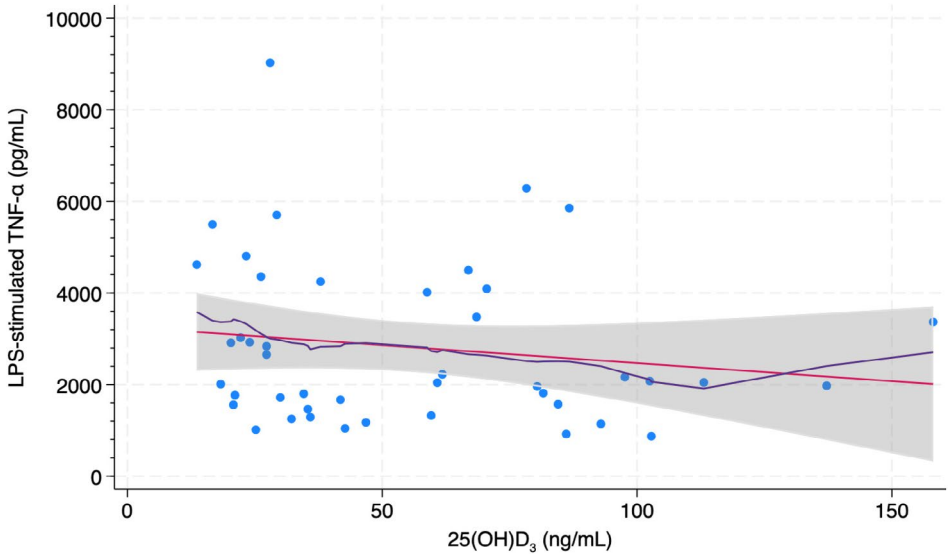

FIGURE 5 | Scatterplot of lipopolysaccharide (LPS)-stimulated tumor necrosis factor (TNF)-α by 25-hydroxyvitamin-D_3 [25(OH)D_3] overlaid with a linear best fit line (magenta line) with 95% confidence intervals. Linear fit confirmed with LOWESS (yellow line). Four data points from 11 dogs, for a total of 44 data points.

TABLE 1 | Median [IQR] for cytokines by stimulant.

Cytokine (pg/mL)	Stimulant	Baseline	Placebo	Low-dose	High-dose
IL-6	PBS	49 [49, 49]	49 [49, 49]	49 [49, 49]	49 [49, 49]
	LPS	721 [533, 926]	498 [439, 721]	500 [422, 648]	528 [381, 712]
	LTA	82 [49, 90]	50 [49, 114]	49 [49, 78]	53 [49, 69]
IL-10	PBS	195 [195, 195]	195 [195, 195]	195 [195, 195]	195 [195, 195]
	LPS	1600 [1120, 1851]	1589 [1082, 2125]	1228 [987, 1587]	1447 [1067, 2116]
	LTA	195 [195, 195]	195 [195, 199]	195 [195, 195]	195 [195, 195]
TNF-α	PBS	49 [49, 100]	49 [49, 68]	49 [49, 66]	49 [49, 58]
	LPS	2914 [1770, 4247]	1801 [1465, 4807]	2044 [1326, 3477]	2049 [1143, 4497]
	LTA	107 [50, 236]	151 [77, 203]	73 [60, 167]	146 [85, 182]

Note: Four data points from 11 dogs, for a total of 44 data points.
Abbreviations: IL, interleukin; LPS, lipopolysaccharide; LTA, lipoteichoic acid; PBS, phosphate-buffered saline; TNF, tumor necrosis factor.

5 | Discussion

The vitamin D metabolome has pleiotropic effects that extend beyond calcium, phosphorus, and skeletal homeostasis, especially with regard to its immunomodulatory properties that have garnered increasing interest in dogs after several decades of evidence in human patients [1–3, 14]. Although various in vitro studies have investigated immune responses to vitamin D in dogs, immunological changes after PO vitamin D supplementation in vivo have not been explored previously. We sought to determine whether vitamin D supplementation conveyed a similar anti-inflammatory shift in the cytokine milieu as has been described in in vitro experiments [6, 8, 9]. Indeed, we identified a significant decrease in antigen-stimulated IL-6 production, a pro-inflammatory cytokine, with calcifediol administration. These results highlighted that as serum 25(OH)D_3 concentrations increased, a concomitant decrease occurred in LPS-stimulated IL-6 production. No visual or statistically significant association with IL-10 production was identified. Low-dose and high-dose calcifediol supplementation for 7 days increased serum 25(OH)D_3 concentrations by more than 150% and 300%, respectively, without any reported adverse events or clinically relevant changes in iCa, creatinine, or BUN concentrations.

Low-dose and high-dose calcifediol supplementation resulted in decreased LPS-stimulated IL-6 concentrations compared with baseline. However, no differences were observed with placebo administration compared to low- and high-dose calcifediol administration. This anti-inflammatory shift was apparent in MELR results as IL-6 concentrations decreased by 2 pg/mL for every 1 ng/mL unit increase in serum 25(OH)D_3 concentration. Although LPS-stimulated IL-6 concentrations were not significantly lower with placebo administration compared to baseline, visual assessment of the raw data illustrates a noticeable decrease in supernatant IL-6 concentrations that was somewhat similar in magnitude to vitamin D supplementation. Our data

suggest a carryover effect of PO vitamin D supplementation on the placebo group, which likely influenced the results despite efforts to minimize it using statistical adjustment. The majority of the cohort likely was affected by this carryover effect because only three dogs were randomized to receive the placebo during the first phase of the study. The 28-day washout period appeared to be insufficient because serum 25(OH)D_3 concentrations after both washout periods remained significantly higher than baseline. The inverse association between LPS-stimulated IL-6 and serum 25(OH)D_3 concentrations, and the significant increase in 25(OH)D_3 after PO vitamin D supplementation support a likely causal relationship between vitamin D and suppression of antigen-stimulated IL-6 production, independent of group comparisons. However, additional studies that utilize a longer washout period are needed to verify this hypothesis.

Although visual inspection of the scatterplot for LPS-stimulated TNF-α suggested a similar relationship as for IL-6, the association was not significant. These results support our hypothesis that PO calcifediol supplementation would have anti-inflammatory effects on cytokine production. We had anticipated a more robust decrease in TNF-α production based on findings from in vitro studies that found that calcitriol had consistent suppressive effects on TNF-α production in healthy and non-healthy dogs and an observational study that found an inverse relationship between serum 25(OH)D_3 concentrations and LPS-stimulated TNF-α production [6–10, 18]. Interestingly, the pro-inflammatory signaling pathways involving p38 mitogen-activated protein kinase and nuclear factor-κβ cause concurrent expression of TNF-α and IL-6 and both induction routes are suppressed with calcitriol [38, 39]. Therefore, it is reasonable to assume that both of these pro-inflammatory cytokines would have been suppressed. The specific reasons for the lack of significant association between antigen-stimulated TNF-α production and 25(OH)D_3 are unknown and beyond the scope of our study. However, we hypothesize that the use of a relatively small and heterogeneous cohort of dogs likely introduced more variability in vitamin D metabolism, the p38 mitogen-activated protein kinase pathway, the nuclear factor-κβ pathway, or some combination of these, thereby increasing the likelihood of type II statistical error.

No visual or statistically significant relationships were identified with any incubation condition for IL-10. Previous in vitro studies have reported variable effects of calcitriol on IL-10 production in dogs [6–10]. Two studies that identified significant increases in IL-10 production after calcitriol exposure in vitro utilized blood samples from critically ill dogs and from healthy dogs that were first primed with endotoxin before antigen-stimulation [7, 8]. We utilized healthy dogs without endotoxin priming of leukocytes. Therefore, PO vitamin D supplementation may have differential effects on IL-10 production in clinical patients.

Oral supplementation with vitamin D, most commonly as vitamin D_3 (cholecalciferol), in healthy people and those with various diseases decreases constitutive and antigen-stimulated concentrations of pro-inflammatory cytokines (IL-6 and TNF-α) and frequently increases production of IL-10, a common anti-inflammatory cytokine [26–29, 40–43]. This shift in inflammatory milieu has direct and indirect downstream immunomodulatory effects that have at least partly contributed to positive outcomes in people with various infectious, inflammatory, autoimmune, and neoplastic disorders who are treated with PO vitamin D as an adjunctive therapy [21–29, 44]. Only two clinical trials have investigated the benefits of PO vitamin D supplementation in dogs [30, 31]. One such study was a randomized, double-blinded, placebo-controlled trial evaluating PO vitamin D_3 (cholecalciferol) supplementation in dogs with atopic dermatitis [30]. Pruritus and skin lesion scores decreased in dogs treated with vitamin D_3, and serum 25(OH)D_3 concentrations were strongly associated with decreases in pruritus scores [30]. The other study that investigated PO vitamin D supplementation in dogs focused on the impact that PO calcifediol had on parathyroid hormone (PTH) and fibroblast growth factor-23 (FGF-23), biomarkers for mineral and bone disorder in dogs with CKD [31]. Calcifediol was well-tolerated by dogs in that study but did not decrease PTH concentrations as anticipated over the 84-day study period. Fibroblast growth factor-23 concentrations increased over the course of the study but without a placebo control it was unclear whether this observation represented natural disease progression or was related to vitamin D administration. Our results combined with previous studies that have evaluated PO vitamin D supplementation in dogs and people suggest that it may be an intriguing adjunctive treatment for disorders caused or perpetuated by dysregulated cytokine production, or in which morbidity is worsened [21–31, 44].

Both low-dose (2.3 μg/kg$^{0.75}$) and high-dose (4.6 μg/kg$^{0.75}$) PO calcifediol supplementation significantly increased serum 25(OH)D_3 concentrations within 7 days of administration. Specifically, serum 25(OH)D_3 concentrations increased from a median of 22.2 ng/mL at baseline to 62 ng/mL (162.4% increase) with low-dose and 93 ng/mL (313.9% increase) with high-dose calcifediol. A previous study found a similar magnitude of 25(OH)D_3 increase after 7 days of PO calcifediol administration using a dose similar to the low dose used in our study [34]. The mean baseline serum 25(OH)D_3 concentration in that study was 31 ng/mL and increased to 70 ng/mL [34]. Another study focused on dogs with CKD and used PO calcifediol at approximately 2.0 μg/kg/day initially, but the frequency of administration subsequently was decreased to 3 times weekly because of concern for the trajectory of the 25(OH)D_3 concentrations and risk for toxicity in the first 4 dogs enrolled in the study [31]. Serum 25(OH)D concentrations increased over baseline by 241% (baseline, 50.2 ng/mL) at the first 28-day examination (171.4 ng/mL) [31]. Our results combined with findings from the aforementioned 2 studies that utilized similar dosing strategies emphasize that serum 25(OH)D_3 concentrations can be increased rapidly and safely using PO calcifediol supplementation in dogs.

No universally accepted normal reference ranges for 25(OH)D concentrations in dogs are available despite some reference laboratories providing reference ranges. One study that included 282 apparently healthy dogs proposed that a physiologic normal range in dogs is 100–120 ng/mL using a chemiluminescence immunoassay [45]. Vitamin D sufficiency was defined in that study by the 25(OH)D concentration range at the inflection point of parathyroid hormone (PTH), beyond which a minimal additional decrease in PTH was observed. This approach also is commonly used in human medicine but simply identifies the 25(OH)D concentration that optimizes calcium and phosphorus homeostasis and bone health and cannot be extrapolated to

be representative of vitamin D concentrations needed for other health outcomes. One example comes from the observation that 25(OH)D concentrations between 30 and 40 ng/mL decrease the risk for colorectal cancer in human patients [46]. Another example comes from a recent meta-analysis, which found that the lowest risk of all-cause and cardiovascular mortality in individuals with type 2 diabetes mellitus occurred at a vitamin D concentration of 24 ng/mL. [47] Based on these findings, the authors recommended that people with this endocrine disorder maintain vitamin D concentrations above this threshold. Differences in assay techniques used to quantify 25(OH)D concentrations limit direct comparisons among studies and have somewhat impeded progress in developing ideal threshold cut-points for vitamin D in domestic animals. Chromatographic methods, including the HPLC assay used in our study, are currently the most commonly used methods and remain the reference standard in human medicine [48]. Prospective studies that include large and diverse cohorts are needed to determine whether increasing vitamin D concentrations by PO supplementation in dogs can have beneficial impacts on immune function and clinical outcomes.

Our study had several limitations. First, our small sample population may have impeded our ability to identify significant differences or associations (type II error). The small population also had limited genetic diversity that may have affected vitamin D metabolism and immune responses. Second, ours was a short-term experimental study primarily designed to determine the effect of $25(OH)D_3$ repletion on antigen-stimulated cytokine production and tolerance of PO calcifediol supplementation based on changes in BUN, creatinine, iCa concentrations, and clinical adverse effects. Our study was not designed to determine the optimal dose of calcifediol to achieve a particular immune response, nor was it meant to imply that calcifediol would be safe if given long-term. Given the short-term nature of our study, we aimed at determining participant tolerance (i.e., clinical adverse events) and variables expected to be abnormal with acute vitamin D toxicity, including iCa, creatinine, and BUN concentrations. Future studies aimed at evaluating calcifediol dosing recommendations for long-term administration should include a variety of vitamin D metabolites, PTH, FGF-23, renal function parameters, as well as serum calcium and phosphorus concentrations. Another limitation is that our 28-day washout period appeared to be inadequate, as evidenced by serum $25(OH)D_3$ concentrations after washout period 1 (32 ng/mL) and washout period 2 (35 ng/mL), which were significantly higher than baseline. Moreover, $25(OH)D_3$ concentrations were higher with placebo (28 ng/mL) compared to baseline, but the difference was not statistically significant. Our assumption that 28 days would be sufficient was made on the basis of a previously calculated plasma half-life during washout of 1.8 weeks for $25(OH)D_3$ in dogs [34]. In that study, plasma $25(OH)D_3$ concentrations returned to pre-treatment baseline concentrations within 4 weeks of discontinuing supplementation. However, dogs in the previous study were fed a single commercially available dog food that contained no vitamin D [34]. The dogs in our study were maintained on their standard diets, and thus the vitamin D content in those diets may have accounted for a longer than expected washout period. Overall, it is possible that the carryover effects of preceding calcifediol supplementation on $25(OH)D_3$ concentrations may have impacted group comparisons as they pertain to the placebo group. For example, LPS-stimulated IL-6 concentrations were decreased after calcifediol supplementation but were no different compared to placebo. Regression results showing an inverse relationship between concentrations of LPS-stimulated IL-6 production and serum $25(OH)D_3$ reaffirm this hypothesis. Lastly, we focused specifically on TNF-α, IL-6, and IL-10 because they have been the most studied cytokines related to vitamin D in dogs and people. Although our approach was purposeful, it was limited in scope in its overall assessment of vitamin D on the cytokine milieu in dogs.

6 | Conclusion

Short-term PO calcifediol supplementation rapidly and safely increased serum $25(OH)D_3$ concentrations, and this repletion was associated with decreased LPS-stimulated IL-6 production. Similar to people and in vitro studies in dogs, PO vitamin D supplementation may have anti-inflammatory effects in dogs.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by Midwestern University Animal Care and Use Committee (protocol 3017). Authors declare human ethics approval was not needed.

Conflicts of Interest

DSM-Firmenich AG LLC donated the HyD used for calcifediol supplementation in this study and measured $25(OH)D_3$ concentrations in compounded capsules free of charge. DSM-Firmenich AG LLC was not involved with data analysis or manuscript preparation.

References

1. V. Müller, R. J. De Boer, S. Bonhoeffer, et al., "An Evolutionary Perspective on the Systems of Adaptive Immunity," *Biological Reviews* 93 (2018): 505–528.

2. A.-S. Vanherwegen, C. Gysemans, and C. Mathieu, "Vitamin D Endocrinology on the Cross-Road Between Immunity and Metabolism," *Molecular and Cellular Endocrinology* 453 (2017): 52–67.

3. M. Cortes, M. J. Chen, D. L. Stachura, et al., "Developmental Vitamin D Availability Impacts Hematopoietic Stem Cell Production," *Cell Reports* 17 (2016): 458–468.

4. J. Cartwright, A. Gow, E. Milne, et al., "Vitamin D Receptor Expression in Dogs," *Journal of Veterinary Internal Medicine* 32 (2018): 764–774.

5. D. D. Bikle, S. Patzek, and Y. Wang, "Physiologic and Pathophysiologic Roles of Extra Renal CYP27b1: Case Report and Review," *Bone Reports* 8 (2018): 255–267.

6. J. A. Jaffey, J. Amorim, and A. E. DeClue, "Effects of Calcitriol on Phagocytic Function, Toll-Like Receptor 4 Expression, and Cytokine Production of Canine Leukocytes," *American Journal of Veterinary Research* 79 (2018): 1064–1070.

7. J. A. Jaffey, J. Amorim, and A. E. DeClue, "Effects of Calcitriol on Apoptosis, Toll-Like Receptor 4 Expression, and Cytokine Production of Endotoxin-Primed Canine Leukocytes," *American Journal of Veterinary Research* 79 (2018): 1071–1078.

8. J. A. Jaffey, J. Amorim, and A. E. DeClue, "Effect of Calcitriol on in Vitro Whole Blood Cytokine Production in Critically Ill Dogs," *Veterinary Journal* 236 (2018): 31–36.

9. J. A. Jaffey, M. Bessette, Z. Tao, et al., "Effects of Calcitriol on Oxidative Burst, Phagocytic Function, and Leukocyte Cytokine Production in Shelter Dogs," *Canine Medicine and Genetics* 7 (2020): 1–10.

10. J. A. Jaffey, R. Kreisler, T. K. Graves, L. al-Nakkash, R. C. Backus, and L. Allison, "Ex Vivo Immune Function and Modulatory Effects of Calcitriol in Dogs With Naturally Occurring Diabetes Mellitus," *Veterinary Sciences* 11 (2024): 193.

11. J. Rodriguez-Lecompte, A. Yitbarek, T. Cuperus, J. C. Rodriguez-Lecompte, H. Echeverry, and A. van Dijk, "The Immunomodulatory Effect of Vitamin D in Chickens Is Dose-Dependent and Influenced by Calcium and Phosphorus Levels," *Poultry Science* 95 (2016): 2547–2556.

12. T. L. Wherry, R. P. Dassanayake, J. P. Bannantine, T. L. T. Wherry, S. Mooyottu, and J. R. Stabel, "Vitamin D3 Alters Macrophage Phenotype and Endosomal Trafficking Markers in Dairy Cattle Naturally Infected With *Mycobacterium avium* Subsp. Paratuberculosis," *Frontiers in Cellular and Infection Microbiology* 12 (2022): 1021657.

13. N. Tokuda and R. B. Levy, "1, 25-Dihydroxyvitamin D3 Stimulates Phagocytosis but Suppresses HLA-DR and CD13 Antigen Expression in Human Mononuclear Phagocytes," *Proceedings of the Society for Experimental Biology and Medicine* 211 (1996): 244–250.

14. G. Chandra, P. Selvaraj, M. Jawahar, M. S. Jawahar, V. V. Banurekha, and P. R. Narayanan, "Effect of Vitamin D 3 on Phagocytic Potential of Macrophages With Live Mycobacterium Tuberculosis and Lymphoproliferative Response in Pulmonary Tuberculosis," *Journal of Clinical Immunology* 24 (2004): 249–257.

15. B. M. Motlagh, N. A. Ahangaran, and S. M. A. Froushani, "Calcitriol Modulates the Effects of Bone Marrow-Derived Mesenchymal Stem Cells on Macrophage Functions," *Iranian Journal of Basic Medical Sciences* 18 (2015): 672–676.

16. K. Grubczak, D. Lipinska, A. Eljaszewicz, et al., "Vitamin D3 Treatment Decreases Frequencies of CD16-Positive and TNF-α-Secreting Monocytes in Asthmatic Patients," *International Archives of Allergy and Immunology* 166 (2015): 170–176.

17. M. Harishankar, K. Afsal, V. Banurekha, V. V. Banurekha, N. Meenakshi, and P. Selvaraj, "1, 25-Dihydroxy Vitamin D3 Downregulates Pro-Inflammatory Cytokine Response in Pulmonary Tuberculosis," *International Immunopharmacology* 23 (2014): 148–152.

18. L. N. Allison, J. A. Jaffey, N. Bradley-Siemens, Z. Tao, M. Thompson, and R. C. Backus, "Immune Function and Serum Vitamin D in Shelter Dogs: A Case-Control Study," *Veterinary Journal* 261 (2020): 105477.

19. P. Piotti, L. Pierantoni, M. Albertini, and F. Pirrone, "Inflammation and Behavior Changes in Dogs and Cats," *Veterinary Clinics: Small Animal Practice* 54 (2024): 1–16.

20. R. H. Straub and C. Schradin, "Chronic Inflammatory Systemic Diseases: An Evolutionary Trade-Off Between Acutely Beneficial but Chronically Harmful Programs," *Evolution, Medicine, and Public Health* 2016 (2016): 37–51.

21. A. R. Martineau, D. A. Jolliffe, R. L. Hooper, et al., "Vitamin D Supplementation to Prevent Acute Respiratory Tract Infections: Systematic Review and Meta-Analysis of Individual Participant Data," *BMJ (Clinical Research Ed.)* 356 (2017): i6583.

22. S. P. Jørgensen, J. Agnholt, H. Glerup, et al., "Clinical Trial: Vitamin D3 Treatment in Crohn's Disease–a Randomized Double-Blind Placebo-Controlled Study," *Alimentary Pharmacology & Therapeutics* 32 (2010): 377–383.

23. H. M. Pappa, P. D. Mitchell, H. Jiang, et al., "Treatment of Vitamin D Insufficiency in Children and Adolescents With Inflammatory Bowel Disease: A Randomized Clinical Trial Comparing Three Regimens," *Journal of Clinical Endocrinology and Metabolism* 97 (2012): 2134–2142.

24. P. Marckmann, H. Agerskov, S. Thineshkumar, et al., "Randomized Controlled Trial of Cholecalciferol Supplementation in Chronic Kidney Disease Patients With Hypovitaminosis D," *Nephrology, Dialysis, Transplantation* 27 (2012): 3523–3531.

25. A. Mousa, N. Naderpoor, H. Teede, R. Scragg, and B. de Courten, "Vitamin D Supplementation for Improvement of Chronic Low-Grade Inflammation in Patients With Type 2 Diabetes: A Systematic Review and Meta-Analysis of Randomized Controlled Trials," *Nutrition Reviews* 76 (2018): 380–394.

26. S. Sharma, A. Kumar, A. Choudhary, et al., "Neuroprotective Role of Oral Vitamin D Supplementation on Consciousness and Inflammatory Biomarkers in Determining Severity Outcome in Acute Traumatic Brain Injury Patients: A Double-Blind Randomized Clinical Trial," *Clinical Drug Investigation* 40 (2020): 327–334.

27. A. Prtina, N. Rašeta Simović, T. Milivojac, et al., "The Effect of Three-Month Vitamin D Supplementation on the Levels of Homocysteine Metabolism Markers and Inflammatory Cytokines in Sera of Psoriatic Patients," *Biomolecules* 11 (2021): 1865.

28. A. Fassio, D. Gatti, M. Rossini, et al., "Effects on Serum Inflammatory Cytokines of Cholecalciferol Supplementation in Healthy Subjects With Vitamin D Deficiency," *Nutrients* 14 (2022): 4823.

29. P. di Filippo, A. Scaparrotta, D. Rapino, et al., "Vitamin D Supplementation Modulates the Immune System and Improves Atopic Dermatitis in Children," *International Archives of Allergy and Immunology* 166 (2015): 91–96.

30. C. J. Klinger, S. Hobi, C. Johansen, H. J. Koch, K. Weber, and R. S. Mueller, "Vitamin D Shows in Vivo Efficacy in a Placebo-Controlled, Double-Blinded, Randomised Clinical Trial on Canine Atopic Dermatitis," *Veterinary Record* 182 (2018): 406.

31. V. J. Parker, A. J. Rudinsky, J. A. Benedict, A. Beizaei, and D. J. Chew, "Effects of Calcifediol Supplementation on Markers of Chronic Kidney Disease-Mineral and Bone Disorder in Dogs With Chronic Kidney Disease," *Journal of Veterinary Internal Medicine* 34 (2020): 2497–2506.

32. F. Alshahrani and N. Aljohani, "Vitamin D: Deficiency, Sufficiency and Toxicity," *Nutrients* 5 (2013): 3605–3616.

33. M. F. Holick, N. C. Binkley, H. A. Bischoff-Ferrari, et al., "Evaluation, Treatment, and Prevention of Vitamin D Deficiency: An Endocrine Society Clinical Practice Guideline," *Journal of Clinical Endocrinology and Metabolism* 96 (2011): 1911–1930.

34. L. R. Young and R. C. Backus, "Serum 25-Hydroxyvitamin D3 and 24R, 25-Dihydroxyvitamin D3 Concentrations in Adult Dogs Are More Substantially Increased by Oral Supplementation of 25-Hydroxyvitamin D3 Than by Vitamin D3," *Journal of Nutritional Science* 6 (2017): e30.

35. R. A. Kurzbard, R. C. Backus, and S. Yu, "Rapid Improvement in Vitamin D Status With Dietary 25-Hydroxycholecalciferol in Vitamin D Insufficient Dogs," *Journal of Nutritional Science* 10 (2021): e12.

36. J. Jakobsen, H. Maribo, A. Bysted, H. M. Sommer, and O. Hels, "25-Hydroxyvitamin D3 Affects Vitamin D Status Similar to Vitamin D3 in Pigs–but the Meat Produced Has a Lower Content of Vitamin D," *British Journal of Nutrition* 98 (2007): 908–913.

37. R. C. Backus and D. C. Ueda, "Age-Dependent Changes in Plasma Concentrations of 25-Hydroxyvitamin D May Complicate Vitamin D Status Assessment of Immature Cats," *Frontiers in Veterinary Science* 11 (2024): 1365204.

38. M. Stio, M. Martinesi, S. Bruni, et al., "The Vitamin D Analogue TX 527 Blocks NF-κB Activation in Peripheral Blood Mononuclear Cells of Patients With Crohn's Disease," *Journal of Steroid Biochemistry and Molecular Biology* 103 (2007): 51–60.

39. Y. Zhang, D. Y. Leung, B. N. Richers, et al., "Vitamin D Inhibits Monocyte/Macrophage Proinflammatory Cytokine Production by Targeting MAPK Phosphatase-1," *Journal of Immunology* 188 (2012): 2127–2135.

40. D. M. Giraldo, A. Cardona, and S. Urcuqui-Inchima, "High-Dose of Vitamin D Supplement Is Associated With Reduced Susceptibility of Monocyte-Derived Macrophages to Dengue Virus Infection and Pro-Inflammatory Cytokine Production: An Exploratory Study," *Clinica Chimica Acta* 478 (2018): 140–151.

41. R. Grossmann, S. Zughaier, S. Liu, et al., "Impact of Vitamin D Supplementation on Markers of Inflammation in Adults With Cystic Fibrosis Hospitalized for a Pulmonary Exacerbation," *European Journal of Clinical Nutrition* 66 (2012): 1072–1074.

42. L. E. Bartels, M. Bendix, C. L. Hvas, et al., "Oral Vitamin D3 Supplementation Reduces Monocyte-Derived Dendritic Cell Maturation and Cytokine Production in Crohn's Disease Patients," *Inflammopharmacology* 22 (2014): 95–103.

43. H. M. Alsufiani, S. A. AlGhamdi, H. F. AlShaibi, S. O. Khoja, S. F. Saif, and C. Carlberg, "A Single Vitamin D3 Bolus Supplementation Improves Vitamin D Status and Reduces Proinflammatory Cytokines in Healthy Females," *Nutrients* 14 (2022): 3963.

44. P. G. Vaughan-Shaw, L. F. Buijs, J. P. Blackmur, et al., "The Effect of Vitamin D Supplementation on Survival in Patients With Colorectal Cancer: Systematic Review and Meta-Analysis of Randomised Controlled Trials," *British Journal of Cancer* 123 (2020): 1705–1712.

45. K. Selting, C. Sharp, R. Ringold, et al., "Serum 25-Hydroxyvitamin D Concentrations in Dogs–Correlation With Health and Cancer Risk," *Veterinary and Comparative Oncology* 14 (2016): 295–305.

46. M. L. McCullough, E. S. Zoltick, S. J. Weinstein, et al., "Circulating Vitamin D and Colorectal Cancer Risk: An International Pooling Project of 17 Cohorts," *Journal of the National Cancer Institute* 111 (2019): 158–169.

47. A. Jayedi, M. Daneshvar, A. T. Jibril, et al., "Serum 25 (OH) D Concentration, Vitamin D Supplementation, and Risk of Cardiovascular Disease and Mortality in Patients With Type 2 Diabetes or Prediabetes: A Systematic Review and Dose–Response Meta-Analysis," *American Journal of Clinical Nutrition* 118 (2023): 697–707.

48. K. W. Phinney, M. Bedner, S. S.-C. Tai, et al., "Development and Certification of a Standard Reference Material for Vitamin D Metabolites in Human Serum," *Analytical Chemistry* 84 (2012): 956–962.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Scatterplots of serum 25-hydroxyvitamin-D_3[25(OH)D_3] and (A) Phosphate-buffered saline (PBS)-stimulated interleukin (IL)-6, (B) Lipopolysaccharide (LPS)-stimulated IL-6, (C) Lipoteichoic acid (LTA)-stimulated IL-6, (D) PBS-stimulated IL-10, (E) LPS-stimulated IL-10, (F) LTA-stimulated IL-10, (G) PBS-stimulated tumor necrosis factor (TNF)-α, (H) LPS-stimulated TNF-α, and (I) LTA-stimulated TNF-α. Linear fit confirmed with LOWESS (yellow line). Four data points from 11 dogs, for a total of 44 data points. **Figure S2:** Spaghetti plot illustrating longitudinal changes in lipopolysaccharide (LPS)-stimulated interleukin-6 (IL-6) production from baseline through three treatment phases (Phase 1, Phase 2, and Phase 3). Each line represents an individual subject's IL-6 response over time. Enclosed black circles indicate baseline measurements (pre-treatment); red, yellow, and blue circles represent responses during high-dose calcifediol, low-dose calcifediol, and placebo treatment phases, respectively. **Figure S3:** Scatterplots of 25-hydroxyvitamin-D_3 [25(OH)D_3] by (A) ionized calcium, (B) blood urea nitrogen (BUN), and (C) creatinine overlaid with LOWESS. Four data points from 11 dogs, for a total of 44 data points. **Table S1:** Median (IQR) for ionized calcium (iCa), blood urea nitrogen (BUN), and creatinine values by treatment group. Four data points from 11 dogs, for a total of 44 data points.